



NEPAL ENGINEERING COLLEGE

Affiliated to Pokhara University | Department of Computer Engineering

NeuroScan

A Convolutional Neural Network Based Brain Tumor
Classification System from MRI Images

Individual Course Project Report — Image Processing and Pattern Recognition

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Introduction

NeuroScan is an automated brain tumor classification system that applies digital image processing and Convolutional Neural Network (CNN) based pattern recognition to brain MRI scans, built to support but not replace expert diagnosis.

Built with Python, PyTorch and Streamlit, the system pairs a trained CNN with an interactive web app so a scan can be uploaded, preprocessed, classified, and explained with a confidence score — end to end.

EVALUATED ON

Accuracy · Precision · Recall · F1-score, tested end-to-end from image upload to final prediction.

1

Image Input Module

Accepts a brain MRI scan for analysis

2

Preprocessing Module

Resizing, normalization & augmentation

3

CNN Feature Extraction & Classification

Learns discriminative features directly from image data

4

Output Module

Prediction, confidence score & performance visualizations

Problem Statement

Brain tumors are among the most serious neurological disorders, arising from abnormal, uncontrolled cell growth and causing headaches, seizures, vision loss, and impaired motor coordination.

MRI is the leading imaging technique for detection, but manual interpretation by radiologists is time-consuming, subjective, and prone to fatigue-driven error especially across hundreds of slices.

Distinguishing between tumor types is further complicated by overlapping visual characteristics, motivating an automated, CNN-based classification approach.

WHY IT MATTERS

Early diagnosis directly improves treatment outcomes and patient survival rates.

4

MRI-derived tumor categories studied

100s

of slices reviewed manually per case

CNN

learns features without hand-engineering

Objectives

1

Design and implement a CNN capable of classifying brain MRI images into Healthy and Tumor categories.

2

Apply digital image preprocessing (resizing, normalization, augmentation) to improve model robustness.

3

Evaluate the trained model using accuracy, precision, recall, and F1-score.

4

Develop an interactive, user-friendly web application for uploading MRI images and viewing real-time predictions.

5

Reduce manual analysis workload and increase the speed and consistency of preliminary MRI screening.

Scope of the Project

IN SCOPE

- Classification into four MRI categories: Glioma, Meningioma, Pituitary, No Tumor
- Public Kaggle Brain Tumor MRI dataset (7,153 images)
- Preprocessing: resizing, normalization, data augmentation
- Custom CNN trained from scratch for multi-class classification
- Evaluation via accuracy, precision, recall, F1-score, confusion matrix
- Streamlit web app for upload, prediction & confidence scores

LIMITATIONS

- ✗ Tumor segmentation or localization within the image
- ✗ Estimating tumor size, stage, or severity
- ✗ Treatment recommendations
- ✗ Clinical validation on real hospital data

The system is an academic demonstration of computer-aided diagnosis — not a replacement for professional medical diagnosis.

Literature Review

Classical CAD Methods

Early systems relied on handcrafted features (SIFT, HOG, LBP, GLCM, PCA) with SVM / Random Forest / KNN classifiers — limited generalization.

Rise of CNNs

CNNs automatically learn hierarchical features — edges and textures in early layers, anatomical structures in deeper layers — surpassing handcrafted pipelines.

Transfer Learning

Pretrained networks (VGG16, ResNet50, DenseNet121, InceptionV3, EfficientNet) fine-tuned on medical images reduce training cost and boost accuracy.

Preprocessing & Augmentation

Resizing, normalization, skull stripping, flips and rotations reduce scanner variability and overfitting across datasets.

Evaluation Practice

Beyond accuracy, precision, recall, F1-score, confusion matrices and ROC/AUC are used given the clinical cost of misclassification.

Dataset Description

Source: Kaggle — Brain Tumor MRI Data (tombackert)

Total images: 7,153 labeled MRI scans, resized to 224×224 and normalized

Reframing: Glioma, Meningioma & Pituitary → "Tumor"; No Tumor → "Healthy"

Class (Original)	Mapped Class	Approx. Images
No Tumor	Healthy	≈ 1,600
Glioma	Tumor	≈ 1,600
Meningioma	Tumor	≈ 1,600
Pituitary	Tumor	≈ 1,700

Predicted class

Healthy

↑ 99.4% confidence

Healthy: 99.4%

Tumor: 0.6%

How an MRI Image Becomes a Prediction

1

Image Input

Upload MRI (JPG/PNG) via Streamlit interface

2

Preprocessing

Resize to 224×224, normalize, augment (train time)

3

Feature Extraction

CNN convolution filters capture edges & patterns

4

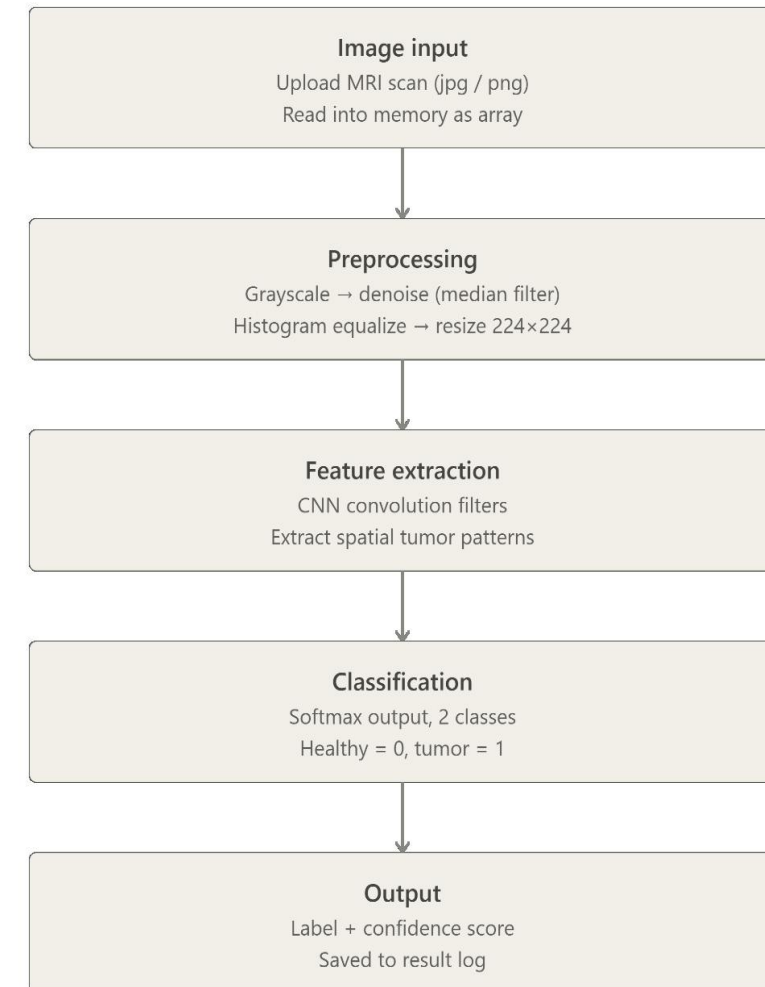
Classification

Fully connected layers output Healthy / Tumor

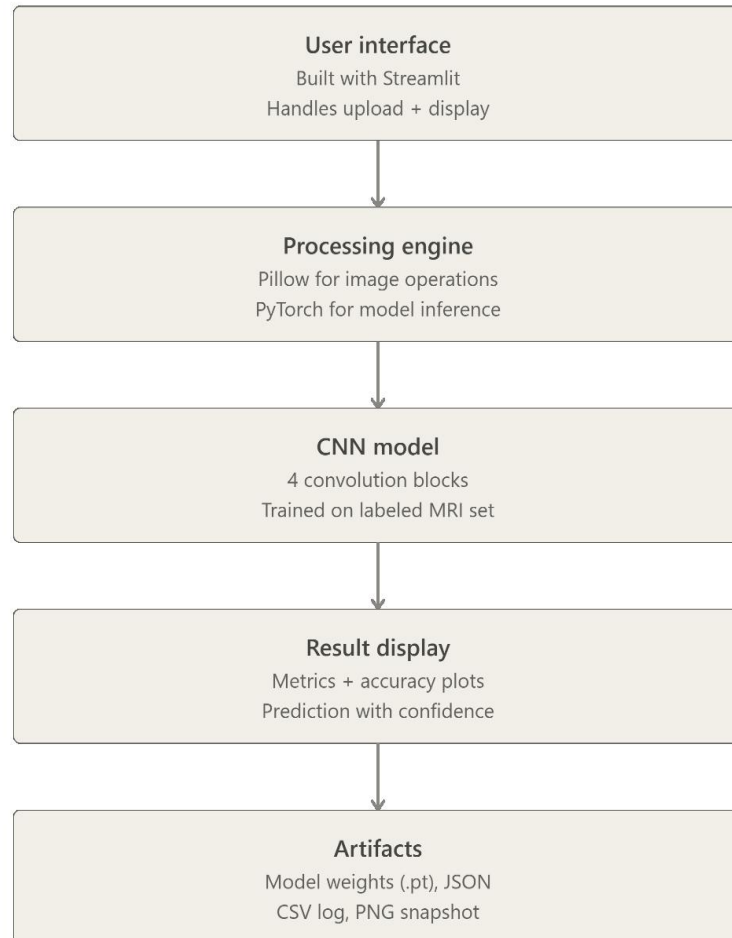
5

Output

Label + confidence score returned to user



System Architecture



User Interface

Streamlit-based UI for image upload & display

Processing Engine

Pipeline for image preprocessing & inference

CNN Model

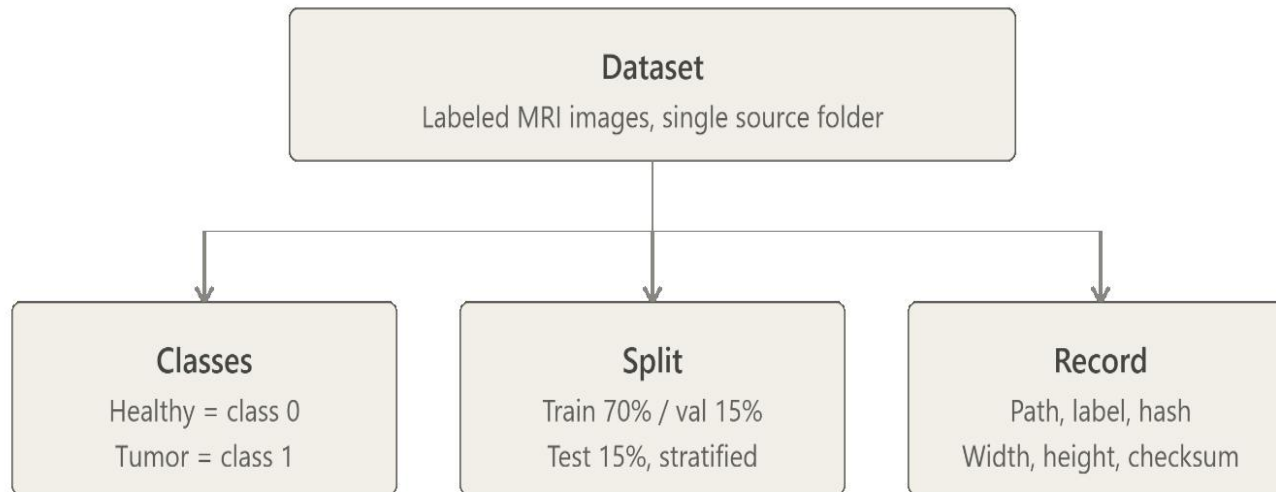
Convolutional blocks trained on labeled MRI set

Result Display

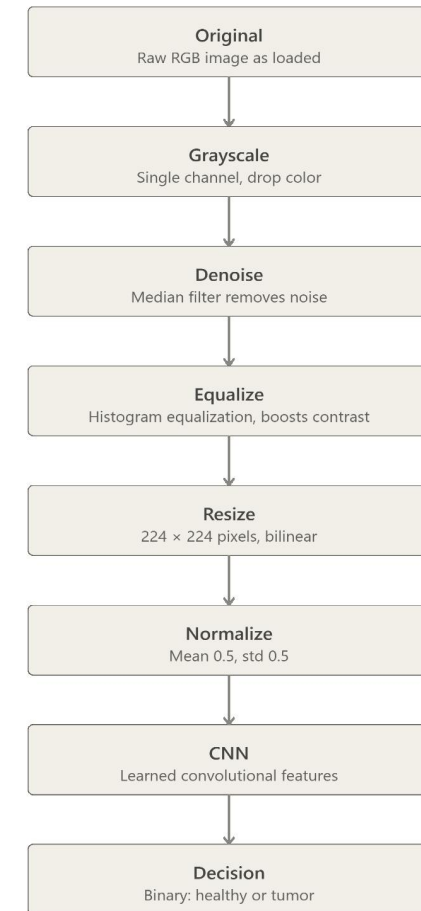
Metrics, accuracy & prediction with confidence

Dataset Organization & Processing Pipeline

Dataset / Class Diagram



Dataset is discovered, split (train/val/test), and passed through resize → normalize → CNN → decision stages.



Processing Pipeline

Seven stages take a raw MRI slice from original pixels through grayscale, denoising, equalization, resizing, and normalization before the CNN makes its healthy/tumor decision.

Tools & Algorithms Used

TOOLS USED

Python 3.13

Base language tying the whole pipeline together

PyTorch

Builds and trains the CNN with GPU acceleration

Torchvision

Data augmentation and image transforms for training

Streamlit

Turns the pipeline into an interactive web app

NumPy / Pandas

Array operations and the dataset manifest

Matplotlib / Seaborn

Training curves and confusion matrix plots

Scikit-learn

Train/val/test splitting and evaluation metrics

ALGORITHMS USED

CNN (BrainTumorCNN)

Learns spatial MRI features; binary tumor classifier

Grad-CAM

Highlights the brain regions behind each prediction

IPPR preprocessing pipeline

Crops, denoises & normalizes scans before the CNN

AdamW + ReduceLROnPlateau

Optimizes weights, decays LR when validation stalls

Stratified grouped split

Balances classes and prevents scan leakage across splits

CNN Architecture & Training Setup

01 Convolution

Learnable filters slide across the image to generate feature maps capturing edges, textures & shapes.

02 ReLU Activation

Introduces non-linearity after each convolution, avoiding vanishing gradients.

03 Max Pooling

Downsamples spatial dimensions, adding translation invariance and reducing compute.

04 Fully Connected

Flattened features pass through dense layers for global reasoning.

05 Sigmoid / Softmax

Outputs a normalized probability for Healthy vs. Tumor classification.

Training uses Binary Cross-Entropy Loss with mini-batch gradient descent via the Adam optimizer.

TRAINING HYPERPARAMETERS

Hyperparameter	Value Used
Epochs	12
Batch size	32
Image size	224 × 224
Learning rate	0.0005
Optimizer	Adam
Loss function	Cross-Entropy Loss

Functional Testing & Performance

Test Case	Expected Result	Status
Upload a valid MRI image	Image accepted and displayed	Pass
Upload a corrupted/invalid file	System shows an error, does not crash	Pass
Predict on a Healthy sample	Model classifies as Healthy	Pass
Predict on a Tumor sample	Model classifies as Tumor	Pass
View dataset statistics	Correct total image count (7,153) displayed	Pass
Train model from the UI	Training completes and metrics update	Pass
View results page	Confusion matrix and training curves render	Pass

PERFORMANCE TESTING

< 2s

per single-image prediction on CPU-only hardware

Robustness verified by testing images from each of the four original classes (glioma, meningioma, pituitary, no-tumor) after mapping to the binary scheme.

Model Evaluation Metrics

0.976

ACCURACY

0.987

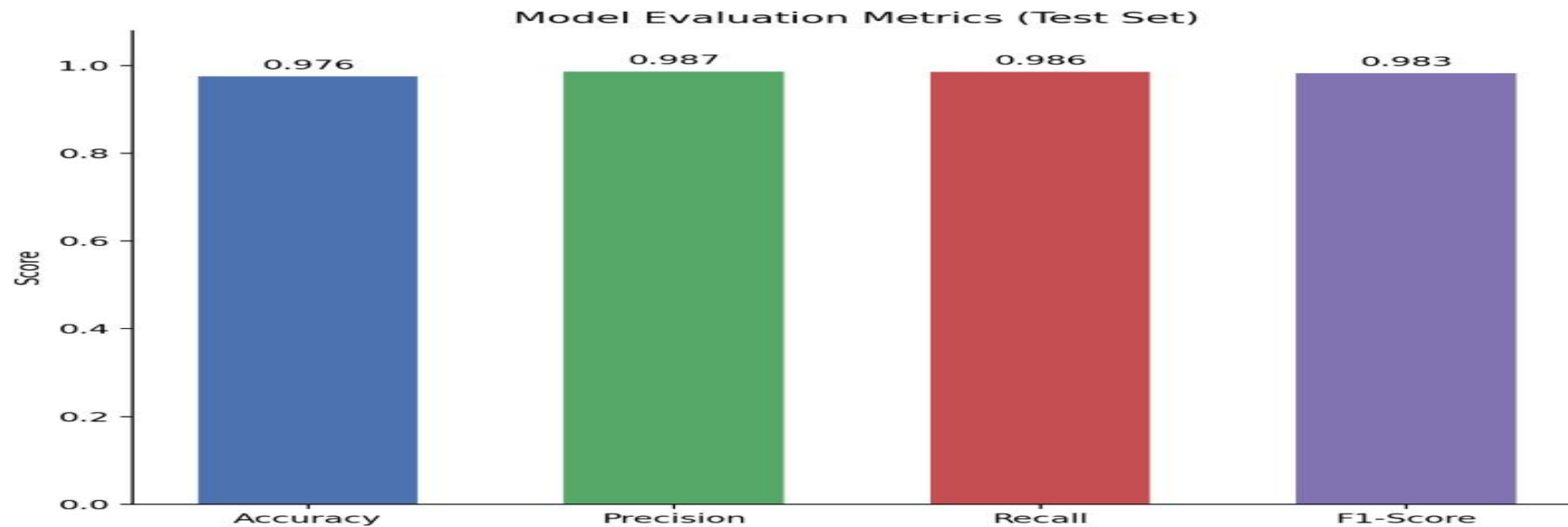
PRECISION

0.986

RECALL

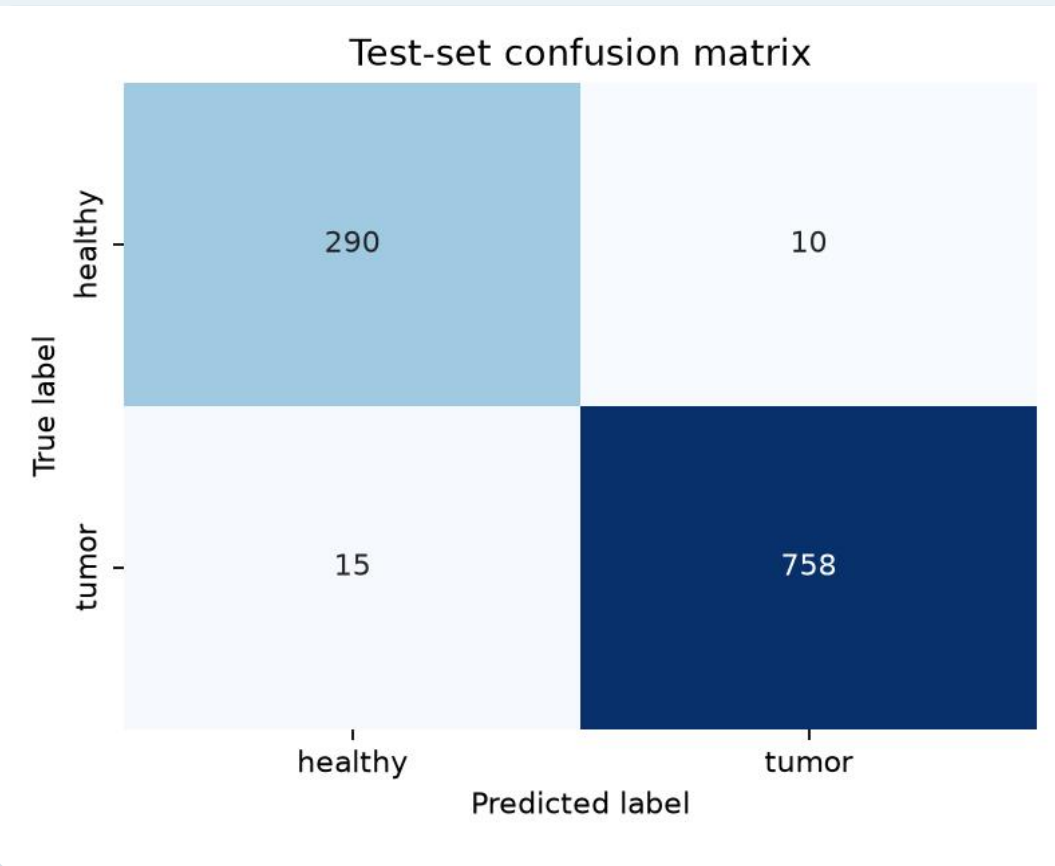
0.983

F1-SCORE

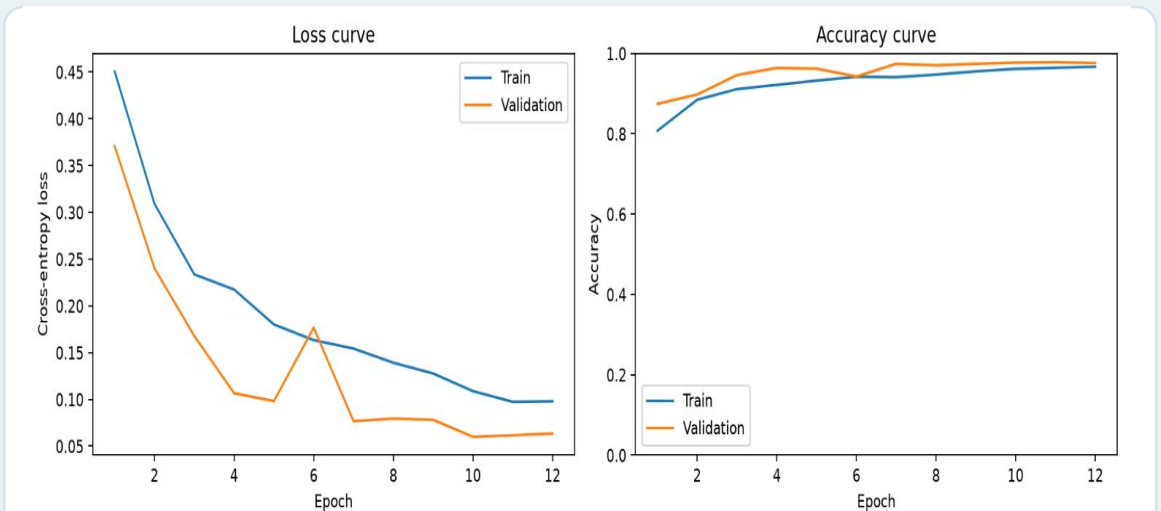


Confusion Matrix & Training Curves

Confusion Matrix



Training & Validation Curves



290 healthy and 758 tumor cases correctly classified, with only 25 total misclassifications on the test set — loss and accuracy curves show stable convergence with no significant overfitting.

Conclusion

● Achievements

- End-to-end brain tumor classification system built successfully
- Streamlit app integrates dataset exploration, training, prediction & results
- Strong test-set performance distinguishing Healthy vs. Tumor scans

● Limitations

- Limited to 2D MRI slices; may not generalize across scanners
- Binary classification only — no tumor sub-typing or localization
- CPU-only training limited experimentation; no clinical validation

● Future Scope

- Extend to multi-class classification (glioma / meningioma / pituitary)
- Add segmentation/localization (e.g. U-Net) to highlight tumor region
- GPU deployment and validation on larger, multi-institutional data

Thank You

NeuroScan — CNN-Based Brain Tumor Classification from MRI Images

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